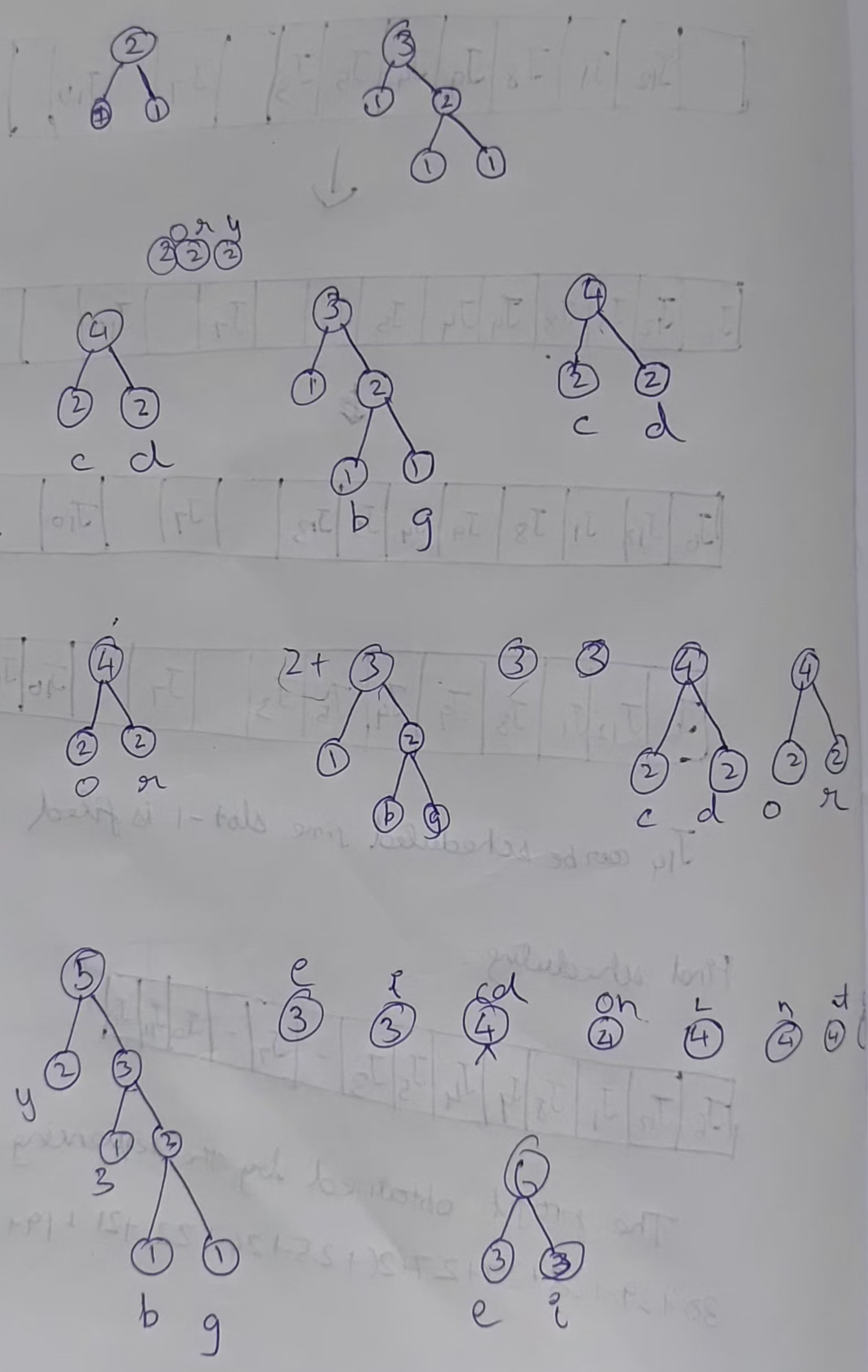
Huffman Coding

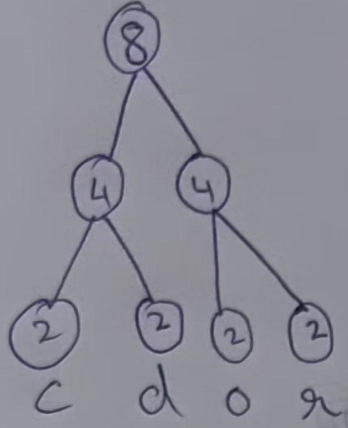
"Data Analytics and Intelligence Laboratory"

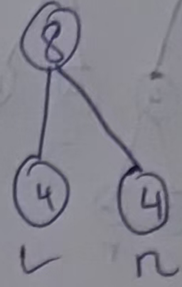
b	g	s	c	d	o	r	y	e	i	l	n	t
1	1	1	2	2	2	2	2	3	3	4	4	4



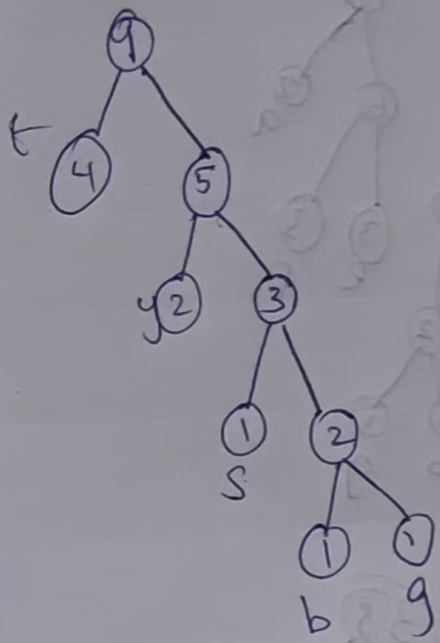
cd	oh	l	n	t	ysbg	ei	a
4	4	4	4	4	5	6	7

t	ysbg	ei	a	cdor
4	5	6	7	8

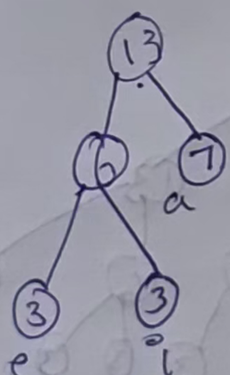




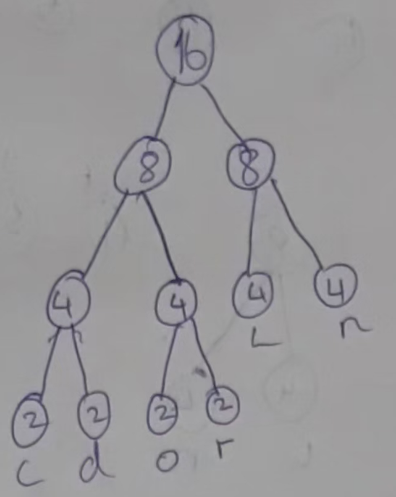
ei	a	cdor	ln
6	7	8	8

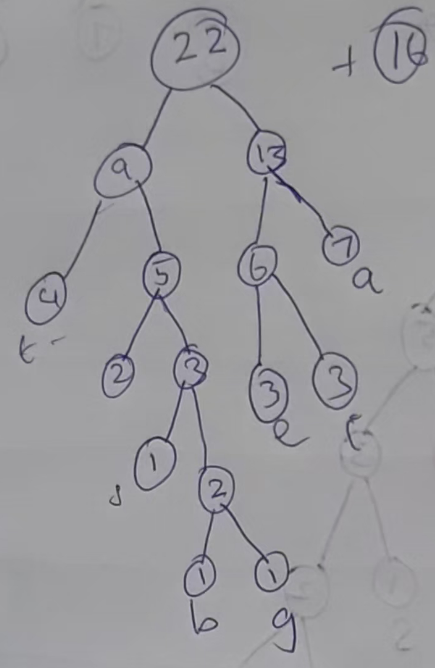


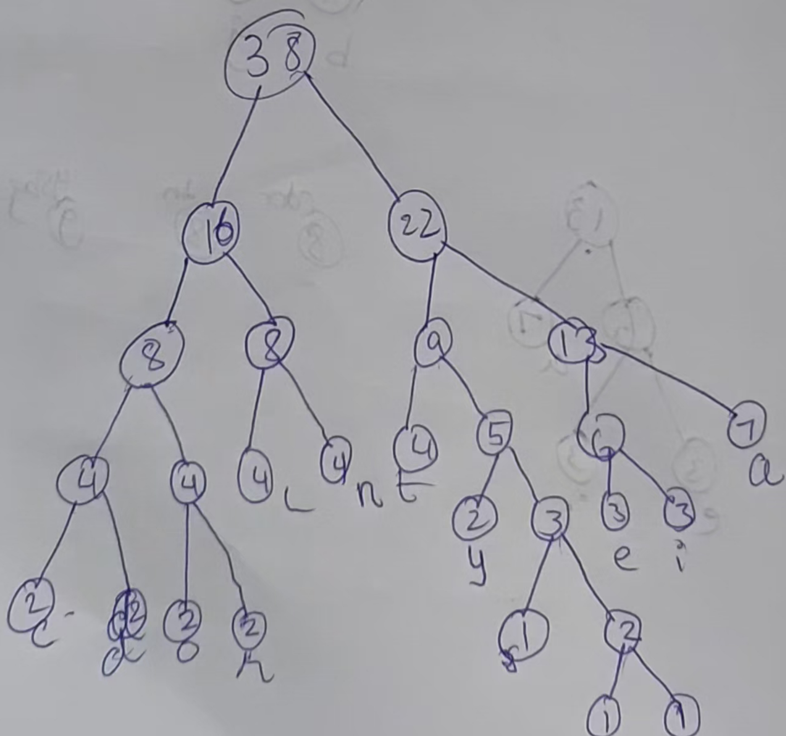
cdor	ln	ysbg
8	8	9



9 ysdg 13 eo







Char	Freq	Code
A	7	111
B	1	101110
C	2	0000
D	2	0001
E	3	1100
G	1	11111
I	3	1101
L	4	010
N	4	011
O	2	0010
R	2	0011
S	4	10110
T	4	100
V	2	1010

Length of Message $= \sum_{i=0}^{n} F_i C_i$

$= 7\times3 + 1\times6 + 2\times4 + 2\times4 + 3\times4 + 1\times5 + 3\times4 +$

$4\times3 + 4\times3 + 4\times3 + 2\times4 + 2\times4 + 1\times5 + 4\times3 +$

2×4

$= 138$ bits